



Class 2: Spectral representation and signal windowing

Group 3: The Smarties

17.05.2025

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✓ Class 2: Spectral Representation and Signal Windowing

✓ Preparatory Tasks

1. How is the sampling frequency defined?

Ans. The rate at which a continuous signal is sampled to create a discrete signal is known as the sampling frequency, and it is commonly represented as f_s . It shows how many samples are captured in a second and is given in Hertz (Hz). It is defined as

$$f_s = 1/T_s$$

where:

- f_s is the sampling frequency
- T_s is the sampling period

The Nyquist-Shannon sampling theorem states that the sampling frequency needs to be at least twice the greatest frequency component of the signal in order to reliably and alias-free reconstruct a signal from its samples:

$$f_s \geq 2f_{\max} \quad (1)$$

where, $f_{\max}$ is the maximum frequency of the original signal

2. What happens to the output signal when under/oversampling?

Ans. When oversampling and undersampling occurs in the signal it is very hard to reconstruct the original signal.

- Undersampling: When sampling frequency is less than twice the highest frequency of the signal then undersampling case occurs in the signal

$$f_s < 2f_{\max}$$

When undersampling happens in the signal at that time aliasing effect is shown up. During this issue high-frequency components of the signal are misrepresented as lower frequencies the signal became distorted and reconstruction to original signal is compromised.

- Oversampling: When sampling frequency is greater than Nyquist rate.

$$f_s \gg 2f_{\max}$$

It reduces the quantization noise, helps to improve Signal-to-Noise ratio, no need for analog anti-aliasing filter which reduces the cost of filters. Provide better resolution and allow us to detect signal fast.

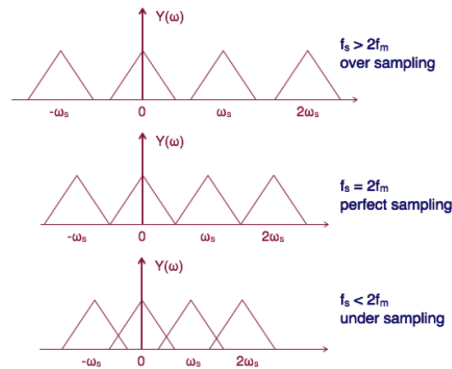


Fig. 1: Sampling cases [1]

3. How do you determine the sampling frequency for a given signal? What happens during superposition of multiple signals?

Ans. Using the Nyquist-Shannon sampling theorem, you must take into account the signal's highest frequency component in order to calculate the sampling frequency. According to this theory, in order to prevent aliasing, the sampling frequency must be at least twice the highest frequency in the signal.

The total of the various signals at each moment in time is the final signal produced when several signals are superimposed. Since the amplitudes of the waves can be directly summed, the superposition principle assumes that the waves are linear. Superposition can occasionally result in constructive or destructive interference, depending on whether the wave amplitudes reinforce or cancel one other out. [1]

4. List a few different windowing functions and explain which function is useful for signal analysis in the frequency domain. What happens to the amplitudes in the time domain?

Ans.

The following are some window functions used in digital signal processing, along with their behavior in the frequency domain.

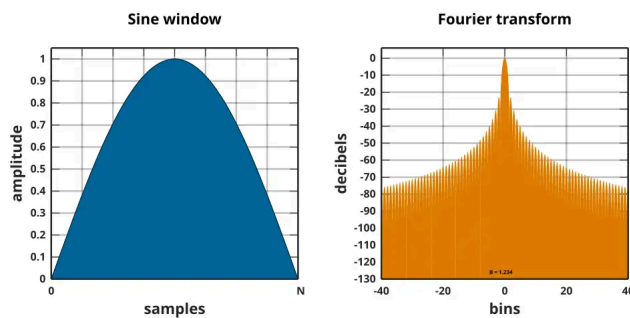


Fig. 2: Sine window [2]

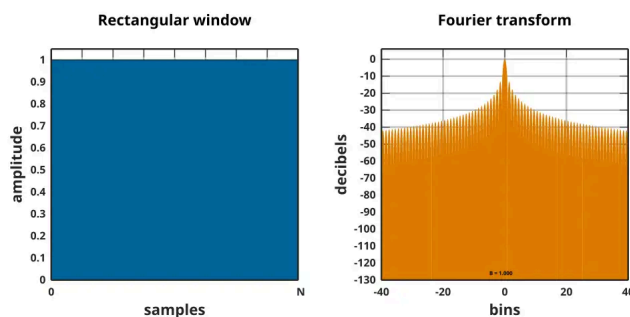


Fig. 3: Rectangular window [2]

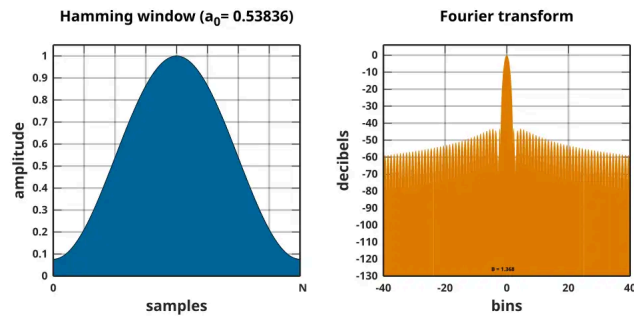


Fig. 4: Hamming window [2]

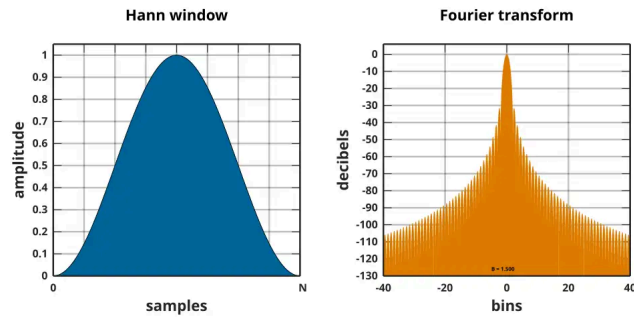


Fig. 5: Hann window [2]

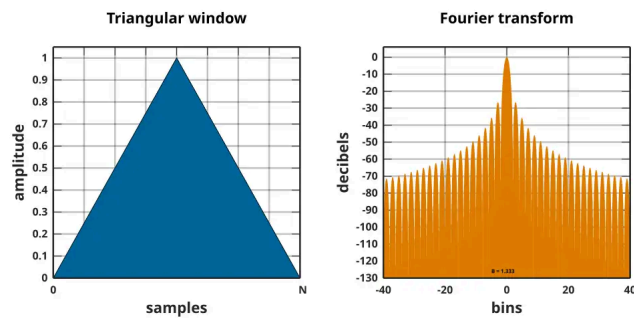


Fig. 6: Triangular window [2]

A window function, sometimes referred to as a smoothing function or tapering function, is a mathematical function that has a zero value outside of a selected interval in signal processing and statistics. Window functions usually taper away from the interval's center, reach a maximum there, and are symmetric about the interval's center. When a window function is mathematically multiplied by another function, waveform, or data sequence, the product is also zero-valued outside the interval the view through the window is the only area that remains. In reality, this means that only the data segment inside the window is multiplied by the window function values after it has been isolated. Therefore, the primary role of window functions is tapering rather than segmentation. [2]

✓ References

[1] GeeksforGeeks contributors. *Sampling in Digital Communication*. 2025. [Online; accessed 16-May-2025].

Available at: <https://www.geeksforgeeks.org/sampling-in-digital-communication/>

[2] Wikipedia contributors. *Window function — Wikipedia, The Free Encyclopedia*. 2025. [Online; accessed 16-May-2025].

Available at: https://en.wikipedia.org/w/index.php?title=Window_function&oldid=1290748431

Exercise 1

Generate signals (sine, rectangle, triangle) and display their spectra individually. Display the spectrum of a sampled signal while varying the sampling frequency. Compare with the original signal.

Exercise 2

Generate a superimposed signal (from 2 sinusoidal signals additively combined) and display it in the time and frequency domains. Sample the signal at three different frequencies and display the sampled values. Compare with the original signal and explain the differences.

Exercise 3

Using the input sinusoid with a frequency of 440 Hz and an amplitude of 1 sampled at 1 kHz, apply three different windowing functions (hamming, triangle, rectangular) and compare the resulting signals in both the time and frequency domains. Explain which window function is useful for identifying amplitude peaks vs. frequency peaks.

Exercise 4

In the given sound file, a car with a honking horn passes a stationary microphone which records the sound. Using the Doppler effect, determine the speed of the car and the frequency of the honking horn. Explain your steps.